

MILESTONE 2

SW Engineering CSC648-848 Spring 2025

Buy-Sell WWW app - GatorMarket

Team 2

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1. **Executive Summary**

San Francisco State University is home to a diverse community of students, faculty, and staff with unique needs for buying and selling goods. Whether it's textbooks, electronics, or furniture, members of the SFSU community often seek affordable items or ways to declutter their belongings. Our product, GatorMarket, aims to bridge this gap by providing an online marketplace exclusively for the SFSU community customized to the needs of this community. This platform will empower users to buy, sell, and coordinate on-campus pickups, fostering both sustainability and community engagement.

GatorMarket will offer an intuitive and user-friendly interface where users can create accounts using their SFSU email, ensuring that only verified members of the community participate. Created by the very students that will be using this product we will be able to implement game changing features such as 'class search' allowing for students to find and purchase class materials specific for their course load. Furthermore the exclusiveness of this product will ensure safety for our users as well as convenience. Products come to you as you can schedule your pick up at designated safe pick up locations on campus. Unlike competitors such as Craigslist or Facebook, GatorMarkets's design is kept simple, with no need for complicated chat features or public postings. This format gives the user exactly what they are looking for, a platform to post and purchase the products they need.

Our team consists of five dedicated computer science majors who understand the unique needs of the SFSU community. With firsthand experience navigating campus life, we are committed to building an accessible, secure, and user-friendly platform. We believe that GatorMarket will become an essential tool for enhancing campus life and fostering a stronger sense of community at SFSU.

2. List of main data items and entities (expand as necessary)

1. User types and permissions

1.1 Registered User:

- **Definition:** SFSU students and faculty have access to the system.
- **Permissions:**
 - Credentials
 - Access to Website
 - View Posts
 - Send and receive messages
 - Post items

1.2 Unregistered User:

- **Definition:** anyone can see the website and interact with items but cannot message/buy or post
- **Permissions:**
 - Create Credentials (become a registered user)
 - Access to Website
 - View Posts

1.3 Admin:

- **Definition:** User to oversee all posts.
- **Permissions:**
 - All registered users permissions
 - Manage the posts and check for appropriateness, approve pending posts

2. Core Entities

2.1 Active Users:

- **Definition:** Active user in session, tracks user authentication
- **Subfields:**
 - activeUserIds

2.2 Message:

- **Definition:** A unit of communication, shown in message history
- **Subfields:**
 - message_id
 - sender_id
 - product_id
 - timestamp

- message_content

2.3 Users:

- **Definition:** displays how many users are on the server. Updated dynamically as users join or leave. Stores authentication details for system access.
- **Subfields:**
 - user_id
 - userEmail
 - userPassword
 - userId

2.4 Filters

- **Definition:** users can search for items. Supports filters and categories for refined searches
- **Subfields:**
 - categories

2.5 Product Listings

- **Definition:** Items posted by users for sale or exchange.
- **Subfields:**
 - product_id
 - owner_id
 - title
 - description
 - price
 - images
 - category
 - Class (optional)

3. SFSU custom entities

3.1 School Locations:

- **Definition:** locations where you can meet up or drop off
- **Subfields:**
 - location_id
 - location_name
 - address
 - description

3. **Functional Requirements - prioritized**

- Priority 1

o Unregistered users

- Unregistered Users shall be able to browse all available listings without an account.
- Unregistered Users shall be able to search for items by title, category, or keywords.
- Unregistered Users shall be able to search items by category and text input.
- Unregistered Users shall be able to view item descriptions, images and item status (available, pending, sold).
- Unregistered Users shall be able to register.
- Unregistered Users shall be able to search for books based on specific SFSU courses (e.g., CSC 648-848) to find textbooks for their classes.

o Registered users

- Users have all of the abilities of Unregistered Users
- Users shall be able to specify a meet up place (e.g., library, student center) for easy trading.
- Only members of the SFSU community shall be allowed to sign up and log in using their university email.
- Users shall be able to create a new listing with a title, description, price, category, and images and post it (subject to admin approval).
- Users shall be able to message the seller.
- Sellers shall have a dashboard showing all their active listings and messages.
- Sellers shall be able to delete an item once a transaction is complete.
- Users shall be able to login.
- Registered Users shall be able to remove their active listings from their dashboard.

o Admin

- Admin shall be required to review and approve an item before becoming public.
- Admin shall be able to delete users.

- Admins shall have the ability to delete listings that violate platform policies

- Priority 2

- o Unregistered user
 - Unregistered Users shall be able to filter items by price
- o Registered user
 -
- o Admin
 - Admins shall have access to an overview of all listings, including reported ones.

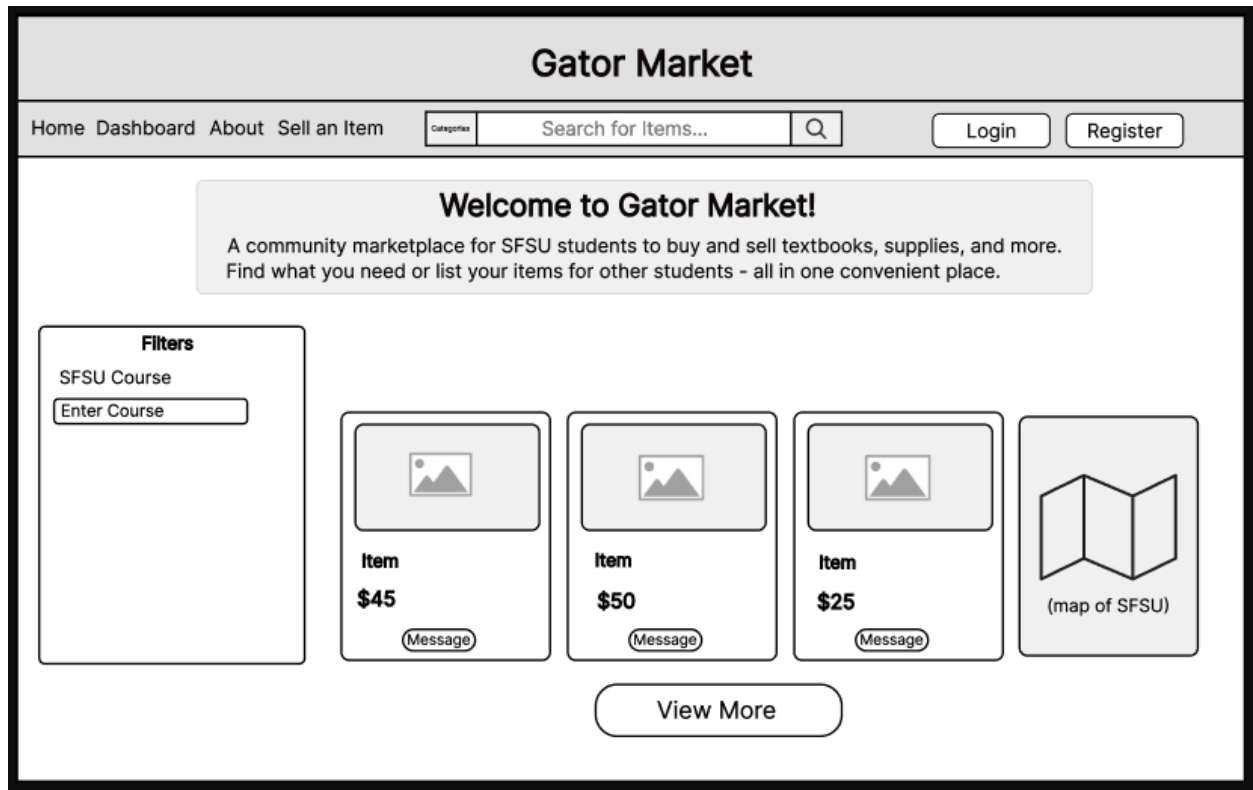
- Priority 3

- o Unregistered user
 -
- o Registered user
 -
- o Admin
 -

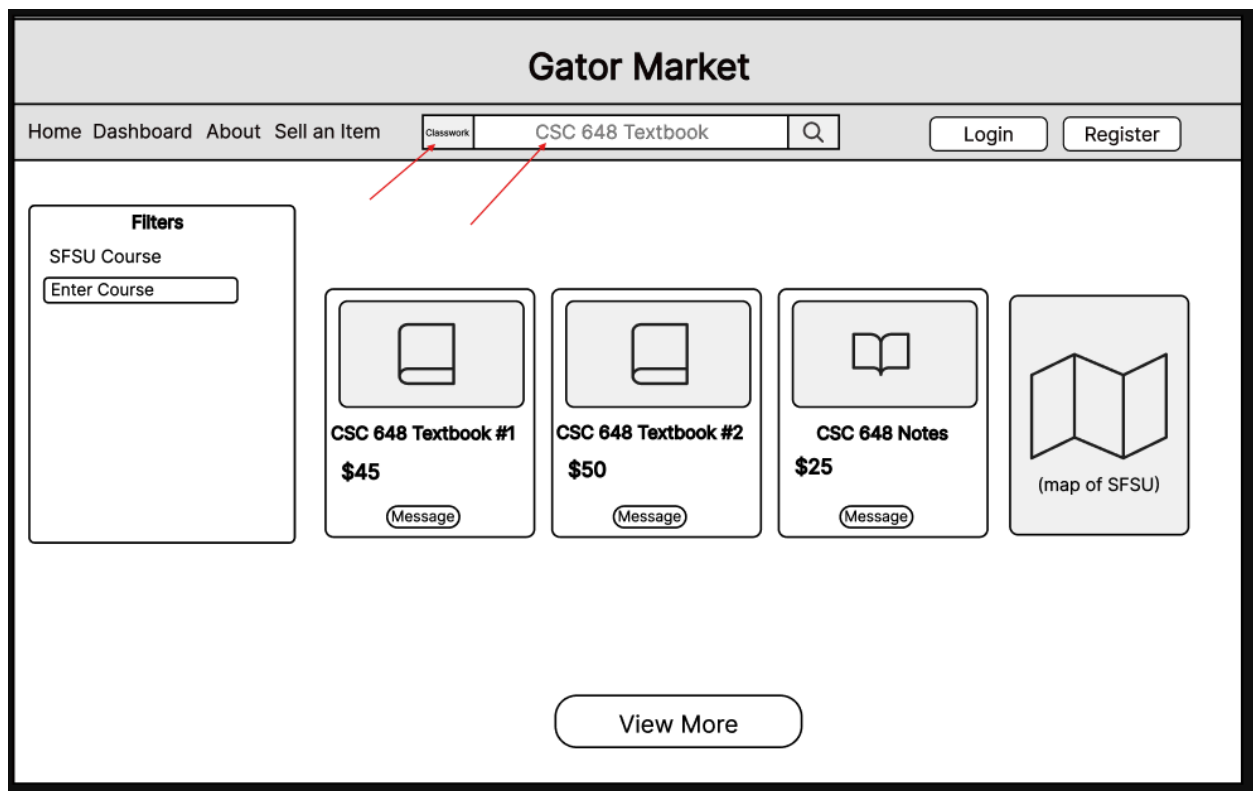
4. UI Storyboards for each main use case

1. Searching and Browsing Listings

Alex is an SFSU student looking for a used textbook, he visits GatorMarket. He is not sure if the book is available, so he browses through the textbooks and course material category and uses the **search** function to **filter** results by title and course name. He browses the items for sale and compares the images and prices before deciding to contact a seller. After typing the **message** and clicking send Alex is prompted to **login** or **register** using his SFSU email. Once in an account he can send the **message** to the seller.



After Searching



Item Page

Gator Market

Home Dashboard About Sell an Item

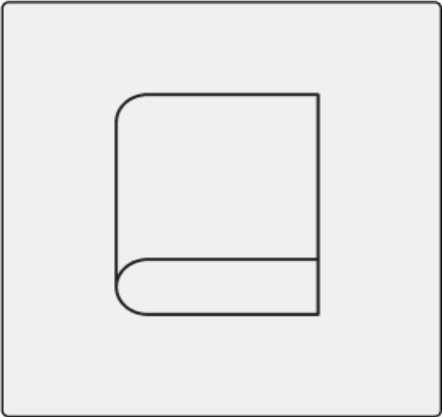
Classwork

CSC 648 Textbook

Q

Login

Register



CSC 648 Textbook

\$45

Category: Textbooks

Course: CSC 648-848

Description:

This is a textbook for csc 648

Message Seller

After Clicking Message Seller

Gator Market

Home Dashboard About Sell an Item

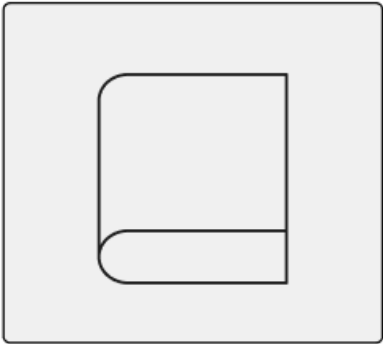
Categories

CSC 648 Textbook

Q

Login

Register



CSC 648 Textbook

\$45

Category: Textbooks

Course: CSC 648-848

Contact Info:

Enter Phone or email for seller to contact you

Message:

Send Message

Filling in Message Box

Gator Market

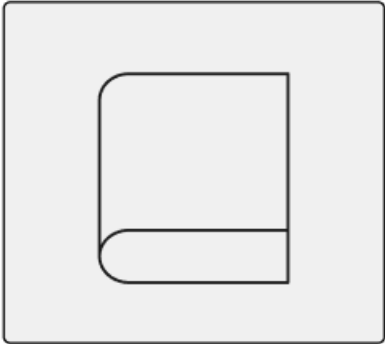
[Home](#) [Dashboard](#) [About](#) [Sell an Item](#)

Categories

CSC 648 Textbook

Q

LoginRegister



CSC 648 Textbook

Contact Info:

alex@sfsu.edu

Message:

Some message about buying the book

Send Message

\$45

Category: Textbooks

Course: CSC 648-848

Sign in Page

Gator Market

[Home](#) [Dashboard](#) [About](#) [Sell an Item](#)

Categories

CSC 648 Textbook

Q

Sign In

Email:

Password:

[Forgot Password?](#)

Sign In

New user? [Register here](#)

Register Page

Gator Market

[Home](#) [Dashboard](#) [About](#) [Sell an Item](#)

Categories

CSC 648 Textbook

Q

Create Account

SFSU Email

Enter email ending in sfsu.edu

Password

Create a password

Confirm Password

Confirm your password

☐ Click to accept [terms](#) and conditions

Already have an account?
Sign in here

Sign Up

After putting in info

Gator Market

[Home](#) [Dashboard](#) [About](#) [Sell an Item](#)

Categories

CSC 648 Textbook

Q

Create Account

SFSU Email

alex@sfsu.edu

Password

Confirm Password

☒ Click to accept [terms](#) and conditions

Sign Up

Back to Message

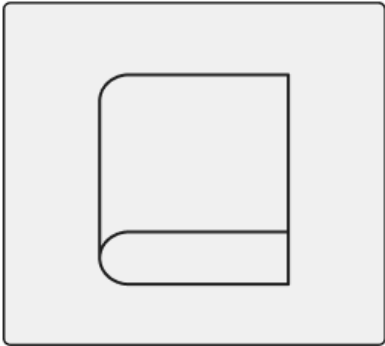
Gator Market

Home Dashboard About Sell an Item

CategoriesCSC 648 Textbook

Q

My Account



\$45

Category: Textbooks

Course: CSC 648-848

CSC 648 Textbook

Contact Info:

alex@sfsu.edu

Message:

Some message about buying the book

Send Message

Message Confirmation

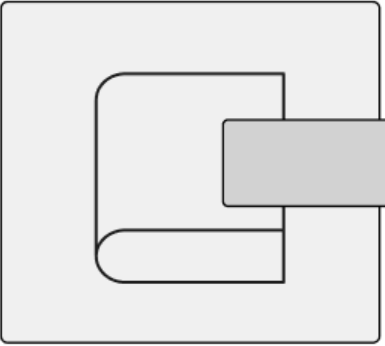
Gator Market

Home Dashboard About Sell an Item

CategoriesCSC 648 Textbook

Q

My Account



\$45

Category: Textbooks

Course: CSC 648-848

CSC 648 Textbook

Contact Info:

alex@sfsu.edu

Message Sent

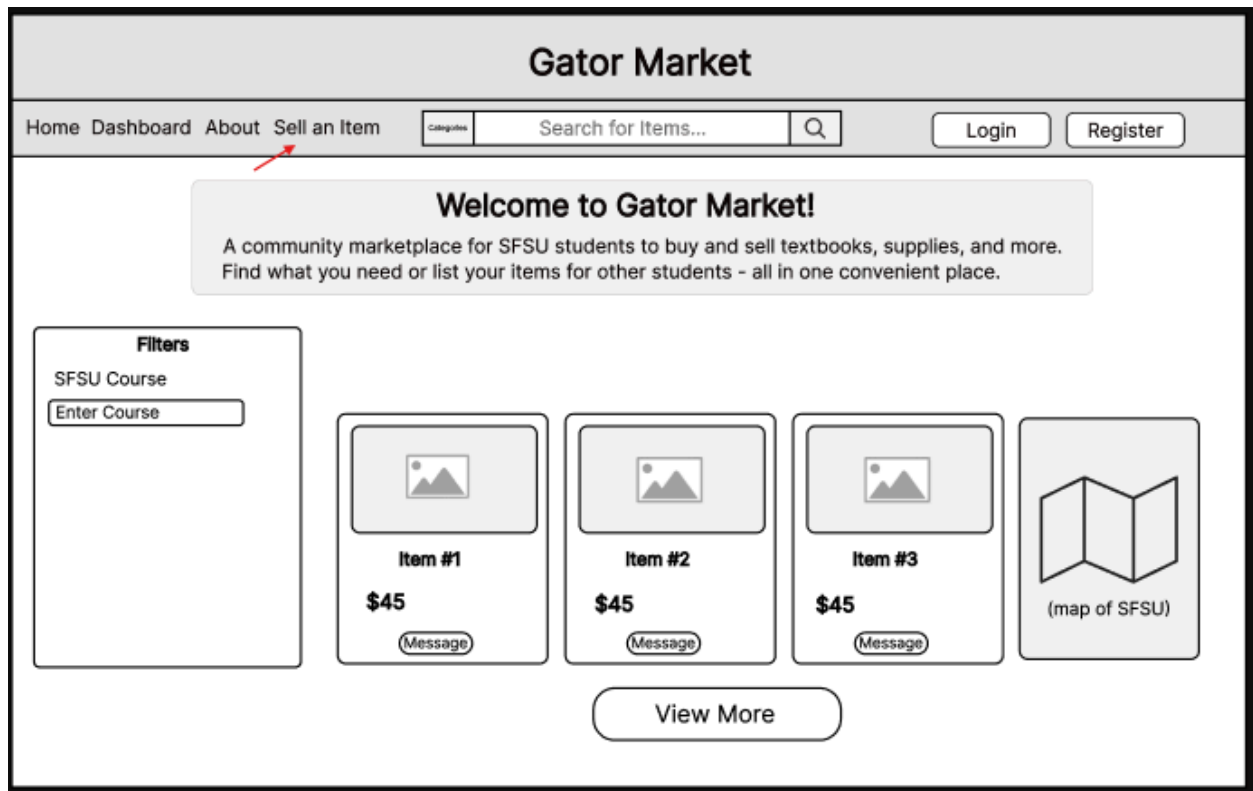
Some message about buying the book

Send Message

2. Posting an Item for Sale

Maria is a graduating senior and wants to sell her old furniture before moving out. She goes to GatorMarket, navigates to the **post** item section, provides details about the furniture, and is prompted to **login** in using her SFSU email when clicking post. She submits her listing and is informed it may take up to 24 hours before it is approved. Once approved, her **post** is visible to other students, and she can wait to be **messaged** by potential buyers.

Home Page



Post Item Page

Gator Market

[Home](#) [Dashboard](#) [About](#) [Sell an Item](#)

Categories

Search for Items...

Q

LoginRegister

Post an Item for Sale

Item Title:

Category:

Price (\$):

SFSU Course (optional):

Description:

Upload Images:

Browse...

Post Item

After Filling in information

Gator Market

[Home](#) [Dashboard](#) [About](#) [Sell an Item](#)

Categories

Search for Items...

Q

LoginRegister

Post an Item for Sale

Item Title:

Sofa

Category:

Furniture

Price (\$):

\$40

SFSU Course (optional):

Description:

This is my old sofa, I am moving out and it is in pretty good condition

Upload Images:

Browse...

Post Item

Login page

Gator Market

[Home](#) [Dashboard](#) [About](#) [Sell an Item](#)

Categories

Search for Items...

Q

Sign In

Email:

Password:

[Forgot Password?](#)

Sign In

[New user?](#) [Register here](#)

After inputting login info

Gator Market

[Home](#) [Dashboard](#) [About](#) [Sell an Item](#)

Categories

Search for Items...

Q

Sign In

Email:

maria@sfsu.edu

Password:

[Forgot Password?](#)

Sign In

[New user?](#) [Register here](#)

Back to post item

Gator Market

[Home](#) [Dashboard](#) [About](#) [Sell an Item](#)

Categories

Search for Items...

Q

[My Account](#)

Post an Item for Sale

Item Title:

Sofa

Category:

Furniture

Price (\$):

\$40

SFSU Course (optional):

Description:

This is my old sofa, I am moving out and it is in pretty good condition

Upload Images:

Browse...

Post Item

Notification

Gator Market

[Home](#) [Dashboard](#) [About](#) [Sell an Item](#)

Categories

Search for Items...

Q

[My Account](#)

Post an Item for Sale

Item Title:

Sofa

Category:

Furniture

Price (\$):

\$40

SFSU Course (optional):

Description:

This is my old sofa, I am moving out and it is in pretty good condition

Upload Images:

Browse...

Post Item

Thank you for making a post!

Allow up to 24 hours for your item to be approved

3. Contacting a Seller

Jordan finds a used laptop for sale and wants to negotiate the price. He clicks on the listing then the **message** button in order to contact the seller through the site's messaging system. The seller receives the **message** in their dashboard. Jordan checks back later for the seller's response and arranges to meet for the transaction. Using the map on GatorMarket that provides safe locations to meet on campus, Jordan finds somewhere to meet the seller. After confirming the **location**, Jordan and the seller meet and successfully complete the transaction.

Home Page

Gator Market

Home Dashboard About Sell an Item

TechnologyLaptops

LoginRegister

Welcome to Gator Market!

A community marketplace for SFSU students to buy and sell textbooks, supplies, and more. Find what you need or list your items for other students - all in one convenient place.

Filters

SFSU Course

Latest Posted



Used Laptop
\$68

Message



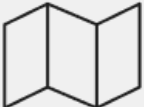
Computer Speaker
\$25

Message



Dorm Chair
\$18

Message



(map of SFSU)

View More

Listing

Gator Market

[Home](#)
[Dashboard](#)
[About](#)
[Sell an Item](#)

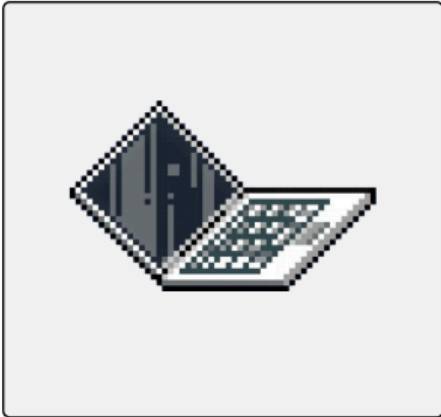
Categories

Laptops

Q

Login

Register



Used Laptop

\$68

Category: Electronics

Course: Any

Description:

The condition is like-new, only used for one semester.
There are no scratches or imperfections to the hardware.
Please feel free to message negotiate before purchasing.
Thank you!

Message Seller

Sellers Dashboard

Gator Market

[Home](#)
[Dashboard](#)
[About](#)
[Sell an Item](#)

Categories

Search for Items...

Q

My Account

Dashboard

+ Sell an Item

Active Listings: 2

Messages: 2

Completed: 7

Recent Messages

Jordan (Used Laptop) - jordan@sfsu.edu
Is this still available? I...

Nick (Dorm Chair) - nick@sfsu.edu
Is this still available? I...

My Listings

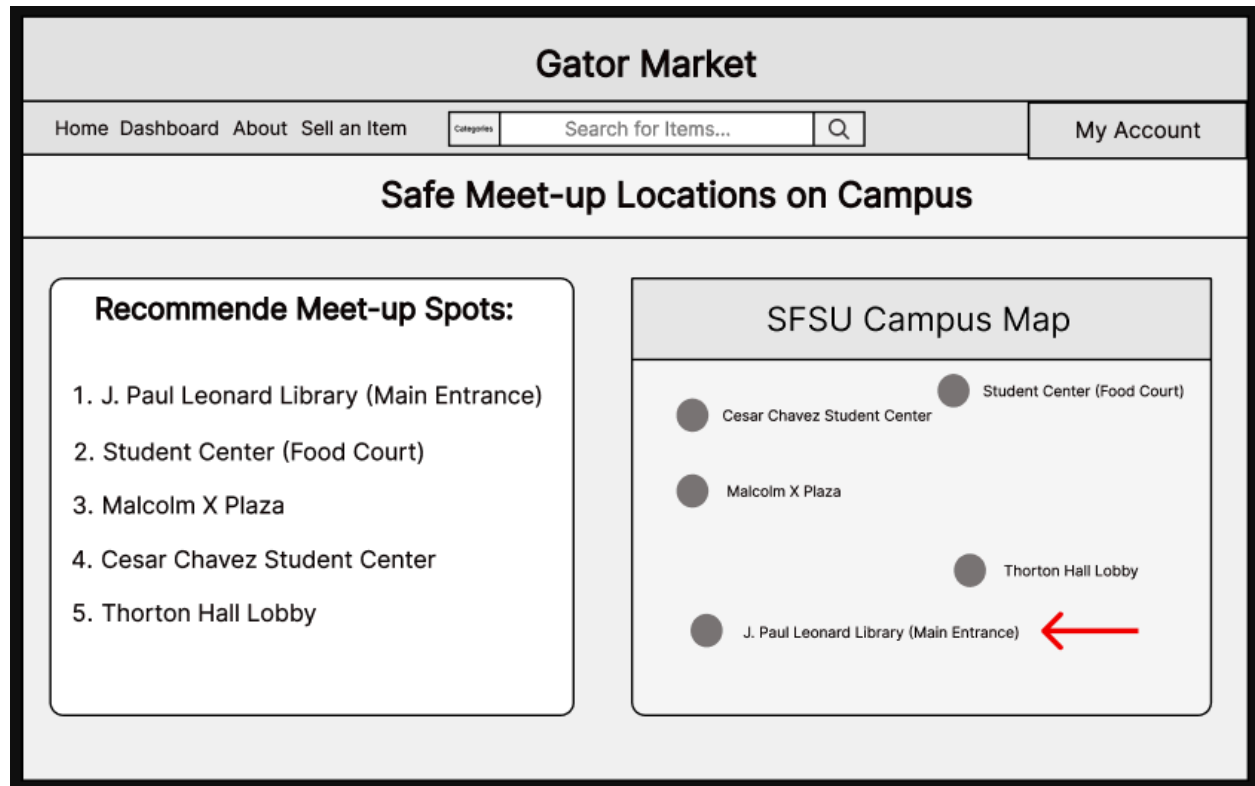
Used Laptop
\$68 - Active - 1 message

Edit

Dorm Chair
\$18 - Active - 1 message

Edit

Meet-up - Seller



4. Admin Review

Lisa is an **admin** for GatorMarket and logs in to review new listings before approving them. She checks for inappropriate content, duplicate listings, or policy violations. When she finds a listing selling an unauthorized service, she denies it and notifies the user.

The admin functions will be performed directly through using Workbench.

5. High-level Architecture, Database Organization summary only

MVC

Model: Handles database interactions and business logic.(user authentication, listing contents, and messaging)

View: Renders the user interface for different user types. (buyer, seller)

Controller: Manages user requests, processes inputs, and communicates between the model and view.

Database Organization

1. Users
 - a. user_id (PK) – Unique identifier
 - b. userEmail – User email address
 - c. userPassword – Encrypted password
 - d. userId
2. ActiveUsers
 - a. user_id (PK) -users currently online
3. Product
 - a. product_id (PK) – Unique listing identifier
 - b. owner_id (FK) – User who posted the listing
 - c. title – Item title
 - d. description – Item description
 - e. category (FK) – Item category
 - f. price – Item price
 - g. class - Associated course (if appropriate)
 - h. Images: file paths to item images
4. Messages
 - a. message_id (PK) – Unique message identifier
 - b. product_id (FK) – Associated product
 - c. sender_id (FK) – User sending the message
 - d. message_content – Message text
 - e. timestamp – Date and time sent
5. Filters
 - a. category (PK) – category for items
6. School_Locations
 - a. location_id (PK) – Unique location identifier
 - b. location_name – Name of the campus location
 - c. address - physical location

- d. description – Additional location information

Media Storage Strategy

Media (images, videos, and audio) will be stored in a file system with references in the database. Relative file paths will be stored in the Product_Listings table to avoid DB performance issues from using BLOBs. File system will be password protected for security.

Search/Filter Implementation

Search Algorithm: SQL with %like statements for title and description searches.

Searchable Columns: title, description, in the Product table. Filters: based on content filtering via the filters table and other fields price or class if extended.

6. Identify actual key risks for your project at this time **schedule risks:**

The first two milestones had a lot of work for the back end team as this is the first step in setting up the environment. As our team for back end is small there are some concerns about the schedule putting stress on them to have a finished product by deadline. Because of this I have been assisting in the back end work to ensure they have enough resources to finish on time.

teamwork risks:

We have found that discord has allowed us to collaborate throughout the week but it puts some stress on the time we have in class to deal with problems in person. While this works for the most part it is dependent on us having class time. As we get further into this project we will have to schedule time out of class for further collaboration.

7. Project management

For Milestone 2 (M2), I ensured that all tasks were clearly assigned with defined deadlines, and each team member was aware of their responsibilities. We've found that discord is a great place for us to collaborate and for me to distribute tasks. I am also able to set up the Milestone document to have all the needed resources the team members will need to complete their assigned work. Including milestone

descriptions, team member assignments, and any class documents they should refer to.

To maintain efficiency, I scheduled mid-point checkpoints to assess progress, identify issues, and adjust timelines as necessary. This ensured that potential issues were addressed before the final delivery. I also emphasized regular communication between the front-end and back-end teams, establishing clear agreements on project structure to enable independent progress while maintaining alignment.

Tools like trello will be helpful for keeping track of tasks for future milestones. As a lot of the current work has been shared on the milestone document. Discord has allowed us to coordinate schedules and google docs for leaving feedback and tracking progress. Discord is also a platform that my entire team is familiar with and will check regularly, however I will be introducing trello as a tool to use to keep everyone on track with their programming tasks as it will better lay out individual goals and deadlines.

8. **Use of GenAI tools like ChatGPT and copilot for Milestone 1**

For this assignment my team attempted to use chatGPT for assistance on their work. Most of my team did not find it very useful as it was for previous milestones. As we are getting more specific to our project and building off previous work and feedback it is harder for chatGPT to achieve the specifics we are looking for.

Our team has a better understanding of the product we are trying to build and is able to articulate it faster than it would take to edit what chatGPT creates.

For this assignment the front end team attempted to use chatGpt to help evaluate the work but since it was mainly image based it was not very helpful. Usefulness: LOW.

The backend team also used prompts of the milestone description and current database structure to evaluate and edit the design. While chatGPT offered feedback it failed to account for the goals of the team and tried to over complicate the data structure. Helpfulness: LOW

ChatGPT was helpful however in creating a description of the high level architecture. Using the description of the database and the requirements of the

section chatGPT was able to generate a description of the media storage and filter processing that coincides with our plan for the database design. Helpfulness: HIGH

9. **Team Lead Checklist to be completed by team lead**

for each item below team lead must answer with only one of the following: DONE/OK; or ON TRACK

(meaning it will be done on time, and no issues perceived); or ISSUE (you have some problems, and then define what is the problem with 1-3 lines)

- So far all team members are fully engaged and attending team sessions when required **DONE**
- Team ready and able to use the chosen back and front end frameworks and those who need to learn are working on learning and practicing **DONE**
- Team reviewed suggested resources before drafting Milestone 2 **DONE**
- Team lead checked Milestone 2 document for quality, completeness, formatting and compliance with instructions before the submission **DONE**
- Team lead ensured that all team members read the final Milestone 2 document and agree/understand it before submission **DONE**
- Team shared and discussed experience with GenAI tools among themselves **DONE**